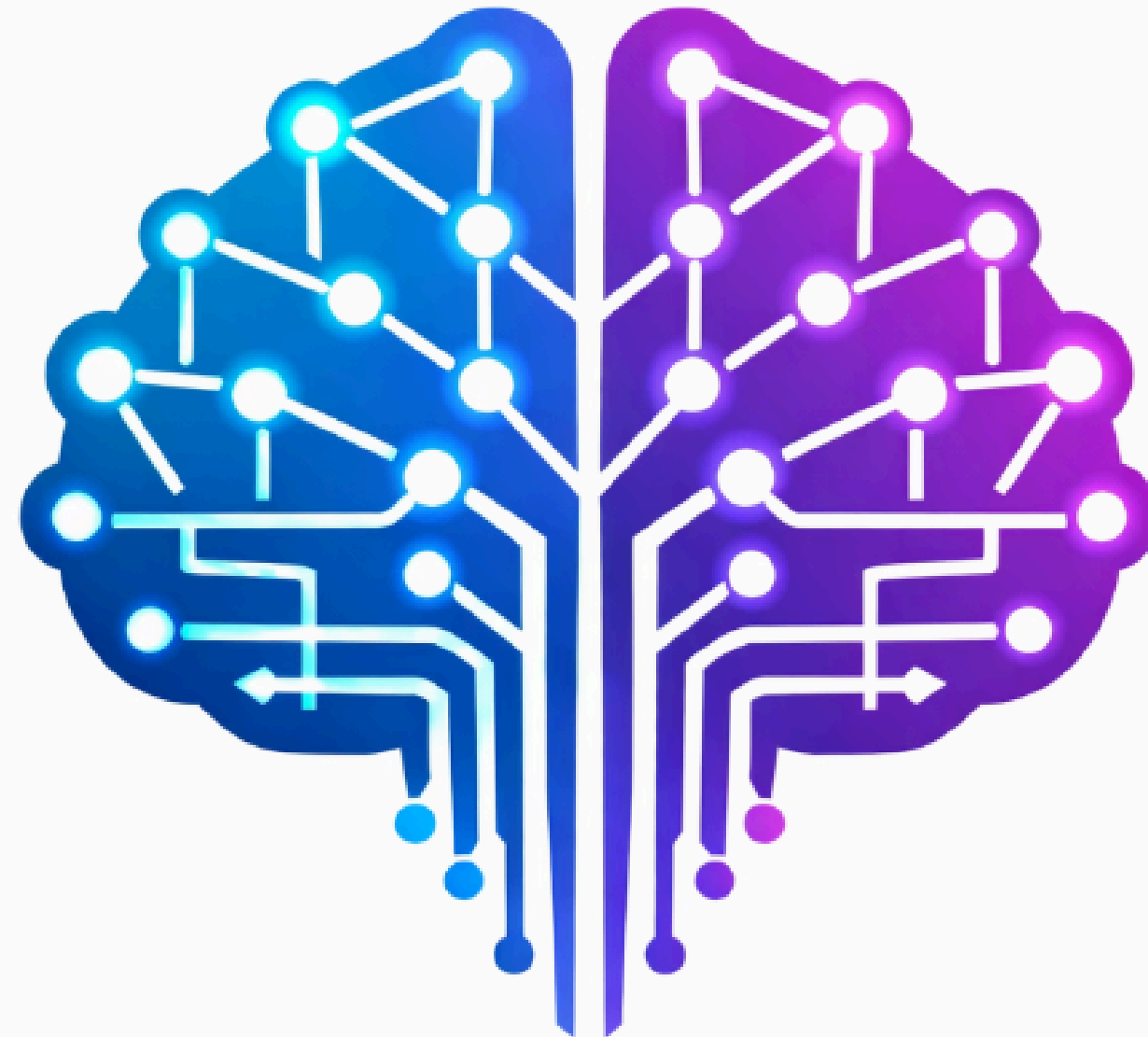


GENAI FUNDAMENTALS



LECTURE:
TOKENS, CONTEXT WINDOWS & COST MECHANICS

Why Developers Must Understand Tokens

LLMs process and **charge usage based on tokens**, not words.

Tokens influence:

- model input size
- response length
- API cost
- context limits

Understanding tokens is essential for building efficient GenAI systems.

What Exactly Is a Token?

A token is a unit of text processed by the model.

Tokens may represent:

- words
- parts of words
- punctuation
- numbers

Example:

"AI is transforming software development."

May become tokens like:

["AI", " is", " transform", "ing", " software", " development", "."]

Input Tokens vs Output Tokens

LLM usage usually involves two types of tokens:

Input Tokens

- prompt
- system instructions
- conversation history

Output Tokens

- generated response

Both are counted for API usage and cost.

What Is a Context Window?

The context window is the maximum number of tokens a model can process at once.

Context includes:

- user prompt
- system instructions
- conversation history
- retrieved documents

Example context windows:

- 8K tokens
- 32K tokens
- 128K tokens

Why Long Conversations Lose Memory

In long conversations, older messages eventually leave the context window.

When this happens:

- earlier information is forgotten
- responses may become inconsistent

Developers must manage context carefully.

How Tokens Affect Cost

Most LLM APIs charge based on token usage.

Cost depends on:

- input tokens
- output tokens
- model pricing

Example scenario:

More tokens → higher cost per request.

Efficient prompt design helps control costs.

Key Takeaway

Tokens and context windows shape how GenAI systems behave.

Developers must manage:

- token usage
- context limits
- prompt length
- response size

Efficient token usage improves **performance, reliability, and cost control.**

